Leading the green economy

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Renewable energy

South Australia is at the vanguard of the global energy transition, lifting net electricity generation from renewable energy from 1% to 74% in just over 16 years.

By 2025/2026, the Australian Energy Market Operator forecasts this could rise to approximately 85%.

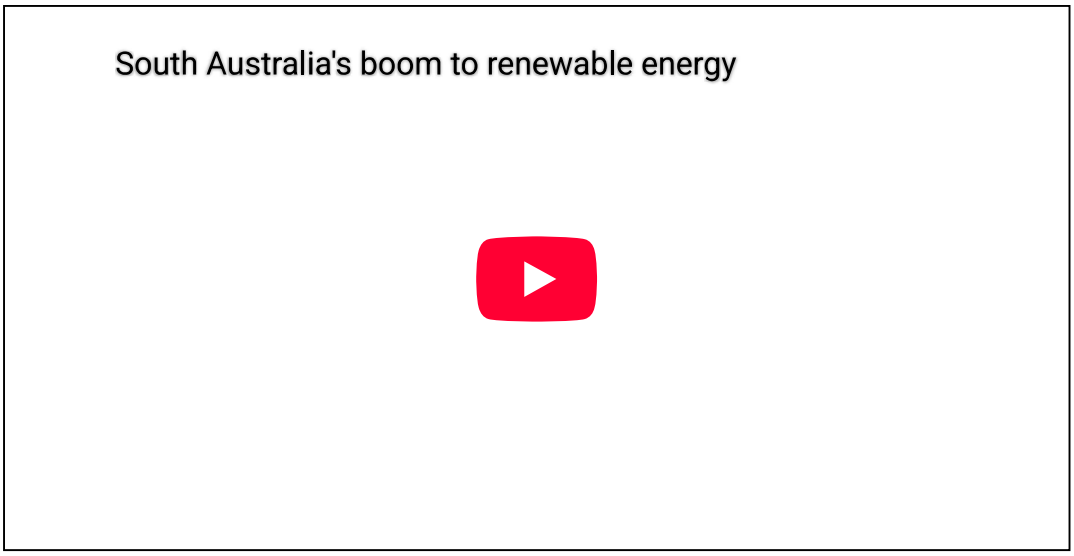
South Australia's aspiration is to achieve 100% net renewables by 2027. In 2021, South Australia met 100% of its operational demand from renewable resources on 180 days (49%).

South Australia has attracted over A$6 billion investment in large-scale renewable energy and storage projects to date with over A$20 billion in the investment pipeline.

We are focused on:

- large-scale renewable energy generation and storage such as wind, solar PV, solar thermal, batteries, pumped hydro, compressed air and thermal storage
- distributed energy resources such as rooftop solar, bioenergy and batteries
- energy efficiency and demand management
- uptake of zero emission vehicles and investment in charging and refuelling infrastructure
- supply-chain development of low-carbon technologies

- research and industry partnerships in low carbon technologies.



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South Australian Government key projects

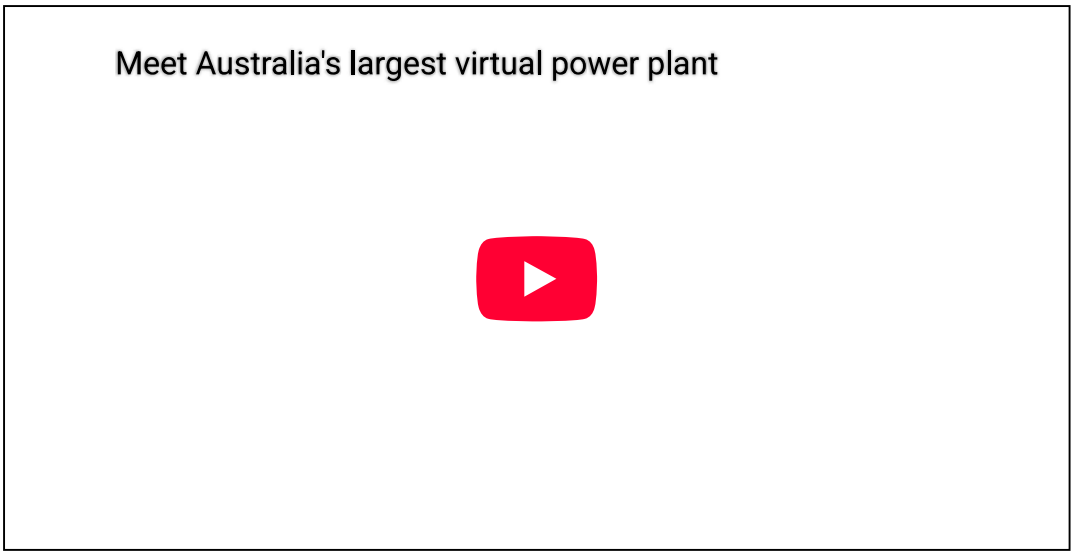
Project EnergyConnect

Project EnergyConnect, a new energy interconnector between South Australia and New South Wales, will support the expansion of the renewable sector by enabling green energy produced in South Australia to be exported to NSW, while providing increased power system reliability.

Learn more about Project EnergyConnect on their website

South Australia's Virtual Power Plant

With the support of the Government of South Australia, Tesla and electricity retailer Energy Locals are developing South Australia's Virtual Power Plant (SA VPP) (https://www.energymining.sa.gov.au/consumers/solar-and-batteries/south-australias-virtual-power-plant) - a network of potentially 50,000 solar and Tesla Powerwall home battery systems across South Australia, all working together to form the world's largest virtual power plant.



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Established industries

Solar

We have one of the best climates for solar, with six large-scale solar farms and one of the highest per capita levels of rooftop solar installed anywhere in the world. Solar PV generated over 24% of South Australia's electricity in 2021/22.

Wind

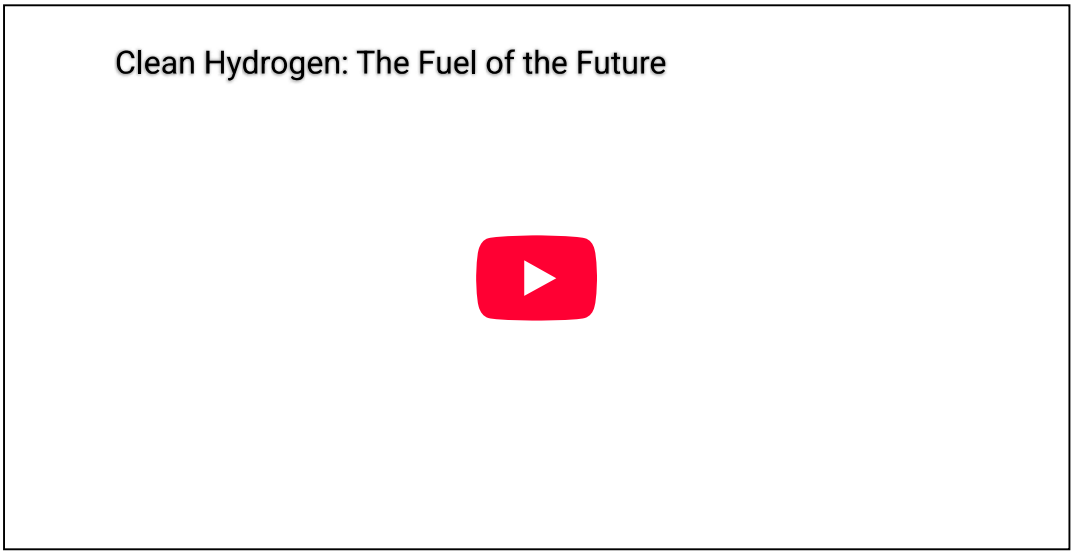
We have world class wind resources, with 24 wind farms in operation. Wind generated over 44% of South Australia's electricity in 2021/22, the highest percentage share of any Australian state or territory.

Battery

South Australia has one of the highest uptake of residential batteries in Australia and has five large-scale batteries in operation and one under construction, headlined by the world-renowned Hornsdale Power Reserve built in 2017 and expanded in 2019.

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Future-ready



Clean hydrogen

South Australia has the wind, sun, land, infrastructure and skills to be a world-class producer, user and exporter of hydrogen.

South Australia's Hydrogen Action Plan sets out 20 key actions across five themes to realise the vision of leveraging the state's competitive advantages to become a world class hydrogen supplier. The key themes are

- Facilitate investments in hydrogen infrastructure
- Establish a world-class regulatory framework
- Deepen trade relationships and supply capabilities
- Foster innovation and workforce skills development
- Integrate hydrogen into our energy system

Read more about the Hydrogen Action Plan (PDF, 12.9 MB) (https://www.energymining.sa.gov.au/industry/hydrogen-and-renewable-energy/hydrogen-in-south-australia/hydrogen-files/south-australias-hydrogen-action-plan-online.pdf)

Read more about the investment potential of South Australia' renewable energy resources: South Australia: a global force in renewable and hydrogen energy (PDF, 2.9 MB) (https://www.energymining.sa.gov.au/__data/assets/pdf_file/0006/891888/205715_SA_A_global_force_brochure_WEB.pdf)

The South Australian Government has to date committed more than $40 million in grants and loans to the development of three megawatt-scale renewable hydrogen projects including:

- **Australian Gas Networks (AGN) – Hydrogen Park South Australia (HypSA)**
  AGN, part of Australian Gas Infrastructure Group (AGIG) recently launched its $14.5 million demonstration project comprising a 1.25 megawatt (MW) electrolyser at the Tonsley Innovation District in Adelaide's southern suburbs. The project is a first step towards decarbonising South Australia's gas network, beginning with 700 homes in the Mitchell Park area now receiving cleaner blended gas, comprising five per cent renewable hydrogen.

- **The Hydrogen Utility™ (H2U) – Eyre Peninsula Gateway**
  Australian hydrogen infrastructure company H2U is developing a world leading $240 million hydrogen project that will see the installation of a 75MW electrolyser near Whyalla, capable of producing enough hydrogen to create 40,000 tonnes of ammonia each year.

In 2020, South Australia completed a pre-feasibility study for exporting hydrogen from South Australia to Asia and produced a hydrogen export modelling tool and prospectus for use by industry.

In 2021, South Australia and the Port of Rotterdam executed a Memorandum of Understanding to study hydrogen export from South Australia to Europe.

Also in 2021, the South Australian government called for Expressions of Interest from industry to access publicly owned land at Port Bonython. Port Bonython was identified through the state's Hydrogen Export Modelling Tool as a highly prospective location for both green and blue hydrogen export.

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Electric vehicles

The South Australian government is making the state's biggest ever investment in electric vehicles, committing $18.3 million to roll out its Electric Vehicle Action Plan (https://www.energymining.sa.gov.au/industry/hydrogen-and-renewable-energy/electric-vehicles).

The government has also announced in October 2021 a $3,000 State Government-funded subsidy for motorists who purchase a new electric vehicle as well as a three year motor registration fee exemption for new electric vehicles purchased before 30 June 2025.

The Electric Vehicle Action Plan includes $13.4 million to leverage private sector investment to accelerate the roll out of a statewide charging network, targeting more than 50 preferred service locations to deliver approximately 500 rapid and fast charging stations.

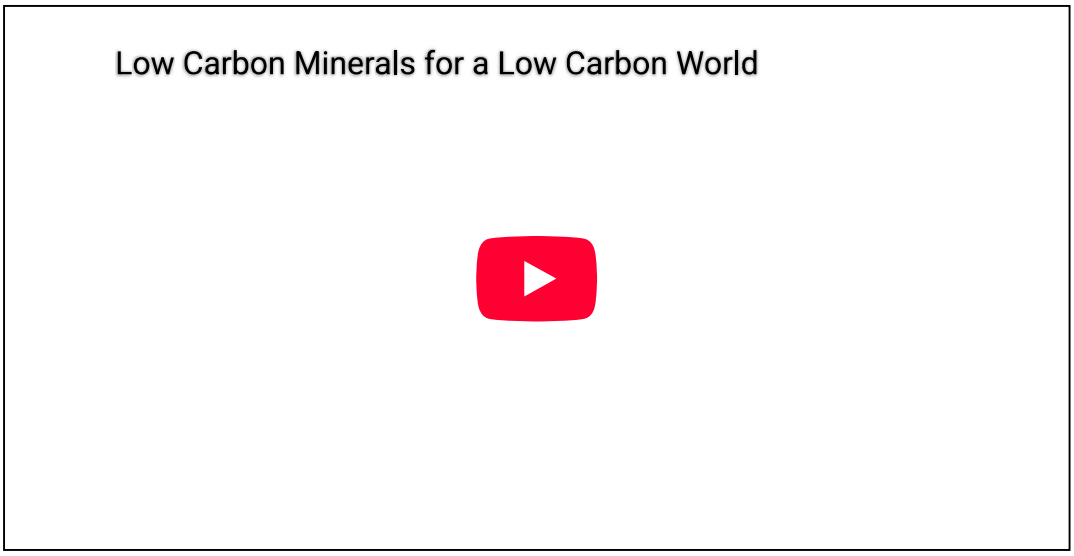
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Clean technology minerals

Minerals are critical to powering the world's clean energy and technology transformation. South Australia's major export – copper – is leading this transition with 67% of Australia's copper resources.

As we move from 60% to 100% net renewable energy by 2030, South Australia is growing 'green' minerals and processing industries to supply copper and other critical minerals across the globe, for low emissions technologies including electric vehicles and renewable energy technology. There is strong demand well into the future for our natural resources, boosted by the world's largest zircon mine, 65% of Australia's graphite, cobalt and magnesite deposits, and new rare earth element discoveries.

Explorers and miners are supported by a modern regulatory system and a willingness to actively pursue a sustainable pipeline of mineral products aiming for low impact activities, zero emissions mines of the future, and sharing the benefits with communities.



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Responding to climate change

South Australia's leadership in transitioning its energy system, abundant natural resources, skilled workforce and research sector provide a strong base from which to transform its economy and achieve its twin goals of reducing state greenhouse gas emissions by more than 50% below 2005 levels by 2030, and achieving net zero emissions by 2050.

South Australia's Climate Change Action Plan 2021–2025 outlines our roadmap to achieving these goals.

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Fast facts

- South Australia covers 1 million square kilometres
- Population: 1.35 million in Adelaide, 1.78 million in South Australia
- South Australia's emissions have reduced by 33% since 2005
- South Australia has committed to achieve net zero emissions by 2050, and has gone one step further to legislate this as well as our goal to reduce net greenhouse gas emissions by more than 50% by 2030
- South Australia has moved from 1% to **over 70% renewable energy generation** in just over 20 years
- South Australia is home to the original world's largest battery, the 150 MW Hornsdale Power Reserve
- More than 1 in 3 (40%) homes in South Australia have solar PV systems installed
- In 2020, 30% of all home batteries installed in Australia were installed in South Australia

- There is around A$20 billion invested in the renewable energy project pipeline
- By 2050, South Australia could generate five times more renewable energy than it needs to meet current grid demand
- South Australia is home to the world's largest zircon mine, and 65% of Australia's graphite, cobalt and magnesite deposits
- In 2021, South Australia produced 305,000 tonnes of copper – that's enough copper to power 3.4 million electric vehicles or 70,000 wind turbines

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Contacts

If you are looking to identify opportunities, strategic partnerships and get advice on the South Australian energy sector, contact:

Strategic Policy and Delivery
Department for Energy and Mining
Email: DEM.BusinessServices@sa.gov.au (mailto:DEM.BusinessServices@sa.gov.au)

You can also speak to the Energy and Mining sector specialists .

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